

Fair k -Center Clustering on Massive Social Network Data Streams

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Abstract

As a fundamental technique with many real-world applications, including social network analysis, center-based clustering may inadvertently discriminate against certain populations based on factors such as age, gender, or socioeconomic status, particularly when nodes are associated with sensitive attributes. In this work, we study the problem of *fair k -center clustering* in the streaming setting, which seeks to select representative items from a large data stream while respecting *group-representation fairness*. Given an input dataset in Euclidean space partitioned into m disjoint groups, the *fairness* constraint requires that the number of centers selected from each group satisfies a given upper bound. Moreover, the problem aims to select a set of centers that minimizes the maximum distance from any point to its nearest center (the k -center objective) while satisfying the *fairness* constraint. We present a one-pass streaming algorithm with approximation ratio 4.46, improving the previous best ratio of $(5 + \epsilon)$ for this problem in general metrics. Notably, our result establishes that streaming fair k -center admits a strictly better approximation ratio in Euclidean space than in general metrics, in contrast to the standard k -center problem, whose best-known approximation ratio is 2 in both Euclidean and general metric spaces. Finally, we complement our theoretical results with an empirical evaluation on five real-world social network datasets and million-scale synthetic datasets, demonstrating significant improvements over state-of-the-art methods in clustering quality while maintaining comparable runtime efficiency.

CCS Concepts

• Theory of computation → Approximation algorithms analysis; Facility location and clustering.

Keywords

Algorithmic Fairness; Fairness Clustering; k -Center; Streaming; Social Networks

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Resource Availability:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18369079>.



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1 Introduction

Social media and social network applications have drawn increasing attention due to discriminatory service offerings based on race and gender [29], particularly in representative and decision-making tasks in recent years. When selecting representative points from a biased dataset, the process may produce discriminatory outcomes that reinforce social inequalities in downstream applications [2, 15, 38]. For instance, on social media platforms, a small subset of users (e.g., opinion leaders) often plays a crucial role in shaping social dynamics and influencing behavioral and structural patterns [51]. However, biased selection in such representative or influential-person tasks can lead to discrimination based on attributes such as race, gender, or socioeconomic status. One notable example is provided by Kay et al. [36], who found that Google Image search results displayed a much higher proportion of male CEOs than exist in reality. Motivated by such issues, the concept of representative fairness in the k -center problem has been applied to decision-making tasks across diverse domains [36, 38], including education, finance, and social networks, where ensuring a fair distribution of representative outcomes is essential. Furthermore, Gaddekar et al. [20] introduced the fair k -supplier problem, which addresses cases where certain data points (e.g., low-quality or inappropriate images) must be excluded from consideration. The fair k -supplier formulation restricts representative selections to a specific subset of high-quality data points while simultaneously ensuring fair representation across groups and minimizing the maximum distance from each data point to its nearest cluster.

Clustering is a fundamental unsupervised learning technique used to uncover meaningful structure in data. In social network applications, it can group users with similar (or dissimilar) attributes and identifies representative individuals. In its classical formulation, the k -center problem is a prototypical clustering task that aims to select k representative points, called centers, to minimize the maximum distance from any point to its nearest center. Classical k -center clustering algorithms [23, 28] provide a 2-approximation, and it is $\mathcal{NP}$ -hard to approximate within a factor smaller than 2 for arbitrary metric spaces unless $\mathcal{P} = \mathcal{NP}$ [30]. Furthermore, the Euclidean k -center problem is $\mathcal{NP}$ -hard to approximate within a factor smaller than 1.822 [8], while the current state-of-the-art approximation ratio remains 2.

In this paper, we focus on the representative fair k -center problem in the streaming model, where data points are processed as they arrive and only a small fraction can be stored in limited memory [22]. In this setting, a streaming algorithm must decide, upon the arrival of each data point, whether to store it in memory. With the rapid growth of data generated by mobile devices, major companies such as Google, Apple, Facebook, Yahoo, and Twitter increasingly analyze streaming data to identify useful patterns for improving

user experience [17]. Consequently, substantial research has focused on handling streaming data across a wide range of tasks and applications [7, 15, 35, 42, 49, 53]. Notably, developing a streaming algorithm is more challenging than designing an offline one, as it must make decisions based only on a small portion of the data rather than the entire dataset.

1.1 Our contribution

In this paper, we present a one-pass streaming algorithm for the Euclidean fair k -center problem by using the outsourcing fair k -supplier (OF- k -SUP) problem as an intermediate component. The contributions of this paper can be summarized as follows:

- Via a reduction from Euclidean fair k -center to OF- k -SUP, we obtain a streaming algorithm with approximation ratio 4.464, strictly improving upon the best ratio of $(5 + \epsilon)$ known for streaming fair k -center in general metric spaces.
- We introduce the OF- k -SUP problem and present an approximation algorithm achieving a ratio of 4.464 that leverages maximum matching in general graphs together with tools from computational geometry.
- Extensive experiments on large-scale social network datasets (including general m) demonstrate that our algorithm achieves higher utility than other one-pass streaming methods, while maintaining lower runtime than the two-pass algorithm.

Note that the best ratio for streaming fair k -center in general metrics is $(5 + \epsilon)$, which is tight as established by the counterexample in Fig. 6 in Appendix A. Consequently, our 4.464-approximation in Euclidean space shows that a fair k -center admits strictly better approximability in Euclidean space than in general metrics.

1.2 Related work

In this section, we review the literature on the k -center problem, focusing on studies that incorporate fairness constraints, different distance metrics, and streaming-model settings.

Fair k -center. The fair k -center problem has been extensively studied, with various definitions of fairness proposed and corresponding approximation algorithms, such as representative fairness, which aims to ensure proportional representation of all groups among the selected representatives [14, 20, 32, 38, 40]; group (or colorful) fairness, which ensures that all clusters contain a balanced (approximately equal) number of points from each color or group [1, 3, 6, 9, 13, 31, 54]; and individual fairness, which focuses on ensuring that similar input points are treated (approximately) equally [33, 41, 46, 52] and so on.

Due to space limitations, we primarily focus on the representative fair k -center problem, which constrains the number of centers selected from each class. The matroid center problem with fairness, a generalization of the k -center problem in which the goal is to select centers satisfying a matroid constraint rather than the simple cardinality constraint k , was first addressed by Chen et al. [11], who proposed a 3-approximation algorithm for general metric spaces, albeit with high time complexity $\Omega(n^2 \log n)$. To improve efficiency, Kleindessner et al. [38] formally introduced fair k -center clustering in the context of data summarization, achieving an $O(nkm^2 + km^4)$

running time with an approximation ratio of $O(3 \cdot 2^{m-1} - 1)$. Subsequently, a major improvement was achieved by Jones et al. [32], who reduced the approximation ratio to 3 for general m groups with a running time of $O(nk + n\sqrt{k} \log k)$. Later, Angelidakis et al. [2] proposed a unified framework combining the representative k -center and private k -center formulations, yielding a 15-approximation algorithm with $O(nk^2 + k^5)$ running time. More recently, Chen et al. [12] introduced the diversity-aware fair k -supplier problem, which selects k facilities from a set P consisting of m disjoint groups, with constraints on the maximum number of facilities chosen per group. They presented a 5-approximation algorithm for this problem based on maximum matching. Gaddekar et al. [20] further improved efficiency by proposing a 3-approximation algorithm for the fair k -supplier problem with running time $O((kn + k^2\sqrt{k}) \log n \log k)$.

k -Center under different metrics. The k -center clustering problem, a fundamental task in unsupervised learning, has a long and rich history across different distance metrics, including general metric spaces, Euclidean spaces and others. From a computational perspective, finding the optimal k -center clustering is $\mathcal{NP}$ -hard under any metric [30] unless $\mathcal{P} = \mathcal{NP}$. Moreover, finding a $(2 - \epsilon)$ -approximate solution is also $\mathcal{NP}$ -complete.

In the Euclidean setting, the problem remains $\mathcal{NP}$ -hard to approximate within a factor smaller than 1.822, even in the plane, if one requires the runtime to be polynomial in k [5, 43]. For the approximate version of the metric k -center problem, Gonzalez [23] proposed the farthest-point clustering algorithm, which achieves a 2-approximation ratio with $O(nk)$ running time. Subsequently, Feder and Greene [18] introduced a two-phase algorithm with optimized running time $O(n \log k)$ in Euclidean space. However, no approximation algorithm is currently known to achieve a factor better than 2 for Euclidean k -center.

k -Center clustering in the online streaming model. Incremental clustering was explored by Charikar et al. [10] several decades ago to address the information explosion of hypermedia and the World Wide Web. Their work focused on doubling metrics and achieved an 8-approximation ratio under a single-pass streaming model using $O(k)$ space. Subsequently, Guha [24] proposed a natural one-pass streaming algorithm that attains a $(2 + \epsilon)$ -approximation with space complexity $O(k \log(1/\epsilon)/\epsilon)$ by applying geometric discretization of distances. Special cases of the 1-center and 2-center problems in Euclidean space have also been investigated [27, 37], where better approximation ratios than those for the general k -center problem are achievable in streaming settings.

Recently, for the matroid center problem, Kale [34] introduced a $(17 + \epsilon)$ -approximation one-pass streaming algorithm that stores at most $O(k^2 \log(1/\epsilon)/\epsilon)$ points, where k denotes the rank of the matroid. They also presented an efficient two-pass streaming algorithm achieving a $(3 + \epsilon)$ -approximation. Subsequently, beyond streaming algorithms, distributed approaches for the representative fair k -center problem have been developed for massive-data models [14]. These methods achieve an approximation ratio of $(17 + \epsilon)$ with a running time of $O(kn/l + mk^2l)$ using l processors and m groups. The same work also proposed a two-pass streaming algorithm with a $(3 + \epsilon)$ -approximation for the fair k -center problem. Following this, Lin et al. [40] introduced a one-pass streaming algorithm for fair

k -center clustering that achieves a $(7 + \epsilon)$ -approximation. Building on this line of research, Guo et al. [26] further improved the results, achieving a $(5 + \epsilon)$ -approximation ratio. Building on these studies, our work focuses on the Euclidean setting and aims to improve the approximation ratio in this setting.

2 Preliminaries

Let S be a finite set of n data points distributed in the Euclidean space where $d : S \times S \rightarrow \mathbb{R}_{\geq 0}$ denotes the distance on S . Moreover, $d(s, C) = \min_{c \in C} d(s, c)$ denotes the distance between a point s and the center set C , and $d(S, C) = \max_{s \in S} d(s, C)$. If $d(S, C) \leq \lambda$, we say that the set C covers S within radius λ . For convenience, we use $B_X(s, \lambda) = \{x \mid x \in X, d(s, x) \leq \lambda\}$ to denote the set of points of $x \in X$ within distance λ from s . Then, we use $B_X(\Gamma, \lambda) = \bigcup_{s \in \Gamma} B_X(s, \lambda)$ and $[m] = \{1, 2, \dots, m\}$.

For a given parameter $k \in \mathbb{N}$, the traditional (vanilla) k -center clustering problem is to select a set of k points $C \subseteq S$ such that $d(S, C)$ is minimized. The fair k -center clustering problem generalizes the traditional k -center problem by imposing additional fairness constraints. Let S be partitioned into m disjoint groups $S_1, \dots, S_m$. Given group budgets k_l on each group l and the cardinality constraint $k = \sum_{l=1}^m k_l$, the fair k -center problem is to find a center set C as follows:

$$\begin{aligned} \min_{C \subseteq S} \quad & d(S, C) \\ \text{s.t.} \quad & |C \cap S_l| \leq k_l, \forall l \in [m] \\ & |C \cap S| = \sum_{l=1}^m |C \cap S_l| \leq k. \end{aligned}$$

We denote $C_l = C \cap S_l$ and use analogous notation for other sets (e.g., Λ_l) regarding group l . As the first ingredient of our algorithm, we employ the λ -independent center set as defined below, which has been studied extensively in literature [14, 40]:

Definition 1. (λ -independent center set) We say Γ is a (maximal) λ -independent center set of S , if and only if:

- (1) The distance between any two points $p, q \in \Gamma$ is greater than λ , i.e. $d(p, q) > \lambda$.
- (2) For any $p \in S$, there exists a point $q \in \Gamma$ such that $d(p, q) \leq \lambda$.

Setting $\lambda = 2r^*$, where r^* denotes the optimal cost of k -center clustering, it can be shown that $|\Gamma| \leq k$. Hence, Γ yields a 2-approximation solution for the vanilla k -center problem. However, under fairness constraints, such a solution may be infeasible. Nevertheless, by using this component as a building block, we can construct a streaming algorithm that achieves an approximation ratio of $(1 + 2\sqrt{3} + \epsilon)$. For brevity, we use $\sigma(S, \lambda)$ to denote a λ -independent center set of S .

We transform our problem into the outsourcing fair k -center (OF- k -SUP) problem:

Definition 2. For a set of data points $S = \{s_1, \dots, s_n\}$ and a set of outsourcing points $P = \{p_1, \dots, p_r\}$, both in Euclidean space, a positive integer k denoting the total number of centers (including suppliers), and a center budget τ , the goal is to find a center set $C \subseteq S \cup P$, such that:

- The objective $d(S, C)$ is minimized, i.e., the maximal distance from any point $s \in S$ to its nearest center is minimized.

- Both $|C \cap S| \leq \tau$ and $|C| \leq k$ hold.

When $\tau = k$, the OF- k -SUP problem reduces to the standard k -center problem; when $\tau = 0$, it becomes the classical k -supplier problem. It is well known that the best possible approximation ratios for these problems in metric spaces are 2 for k -center and 3 for k -supplier. In Euclidean space, the k -supplier problem admits a tighter ratio of $1 + \sqrt{3}$. To ensure fairness when selecting centers from both S and P in Euclidean space, we propose an algorithm achieving an approximation ratio of $1 + 2\sqrt{3} + \epsilon$. While this ratio is not tight for the OF- k -SUP problem itself, it allows us to obtain the same bound for the streaming fair k -center problem. For simplicity, and following common practice in the literature [4, 25], we assume that the optimal radius r^* is known throughout the analysis.

3 The Streaming Algorithmic Framework

In this section, we first present the overall algorithmic framework and then focus on the post-streaming selection method, which is the most challenging component and transforms our problem into OF- k -SUP. Although our streaming fair k -center method primarily targets the case $m = 2$ (i.e., two groups as emphasized in [32, 38]), our algorithm also has the potential to extend to the general case of arbitrary m . In addition, we evaluate the practical performance of our approach for general m in the experimental section.

3.1 Algorithmic framework

In essence, our algorithm proceeds in two stages.

- **Streaming stage.** Select m sets of representative data points from the stream; namely, for each group l , compute a λ -independent center set Γ_l with $\lambda = \sqrt{3}r^*$, where r^* is the optimum radius;
- **Post-streaming Stage.** Select points from the family of sets of streamed and stored representative sets $\{\Gamma_l \mid l \in [m]\}$ to form the final set of centers.

For brevity, we use $\Phi = \bigcup_{l \in [m]} \Gamma_l$ throughout the paper. Note that $|\Phi|$ may exceed k , and $|\Gamma_l|$ may exceed k_l for some l . Nevertheless, the following property, which is well known (e.g., textbook [16]), can be used to bound the number of stored points:

Lemma 1. Let O denote an optimal clustering. Any cluster $O \in O$ contains at most two vertices of a λ -independent center set Γ with $\lambda = \sqrt{3}r^*$.

By the above lemma, when $\lambda = \sqrt{3}r^*$, a λ -independent center set contains at most $2k$ data points. Therefore, we have the memory complexity and update time as follows:

Corollary 1. For $\lambda = \sqrt{3}r^*$, the streaming stage requires $O(k)$ memory. Moreover, it takes $O(k)$ update time to compute the distances between the arriving data point of group l and the set Γ_l .

We next establish the following lemma, which states that the union of the streamed and stored λ -independent center sets always contains a feasible center set with a tight performance guarantee (with detailed proof in Appendix A):

Lemma 2. For $\lambda = \sqrt{3}r^*$, there exists a center set $C \subseteq \Phi$ such that:

- For any $u \in \Phi$, $d(u, C) \leq (1 + \sqrt{3})r^*$;
- $|C| \leq k$ and $|C \cap \Gamma_l| \leq k_l$ for each $l \in [m]$.

That is, if we could construct such a set C from Φ , we would immediately obtain the desired solution with an approximation ratio of $1 + 2\sqrt{3}$ by adding the $\lambda = \sqrt{3}r^*$ used in the streaming phase. However, this is a challenging task, because the optimal center set O is unknown and all points except those in Φ have been discarded. In this context, the optimal solution to the fair k -center problem restricted to Φ can have radius $(1 + \sqrt{3})r^*$, since the discarded points can be optimal centers, as illustrated in Fig. 1 in Subsec. 3.4. Nevertheless, we design a selection method that produces a clustering with a min-max radius of $(1 + \sqrt{3})r^*$ regarding Φ , thereby achieving the best radius attainable for Φ itself.

3.2 The post-streaming selection

Our key idea for selecting the center set C from Φ is to iteratively grow a *proper center subset* until the remaining selection problem reduces to an instance of OF- k -SUP. Let $O = \{O_1, \dots, O_k\}$ denote an optimal clustering. The notion of a proper center subset is defined as follows:

Definition 3. (Proper center subset) A set Λ is a proper center subset with respect to Φ if and only if the following conditions hold:

- The uncovered points in $U = \Phi \setminus B(\Lambda, 2r^*)$ appear in at most $k - |\Lambda|$ clusters of O ;
- For these (at most $k - |\Lambda|$) optimal clusters that contain points from U , there exist at most $k - |\Lambda|$ points in Φ , one per cluster, such that:
 - Each of these points covers its corresponding cluster entirely within radius $(1 + \sqrt{3})r^*$;
 - At most $k_l - |\Lambda_l|$ of these clusters are covered by points from group l , where $\Lambda_l = \Lambda \cap S_l$.

We next state the following lemma, which characterizes the subset Λ we aim to grow:

Lemma 3. For a proper center subset Λ with respect to Φ , if $U = \Phi \setminus B(\Lambda, 2r^*)$ contains points from a single group, then the problem reduces to an instance of OF- k -SUP that is feasible regarding radius $(1 + \sqrt{3})r^*$, with cardinality upper bound $k - |\Lambda|$ and fairness constraints $k_l - |\Lambda_l|$ for each group l .

PROOF. Without loss of generality, assume that U contains only points from Γ_1 . Setting $S = U$, $P = \Phi$, and $\tau = k_1 - |\Lambda_1|$ yields an instance of OF- k -SUP. Moreover, by the definition of a proper center subset (see Def. 3), the transformed instance is feasible regarding radius $(1 + \sqrt{3})r^*$. $\square$

Thus, only two tasks remain: (1) grow a proper center subset Λ as in Lem. 3; and (2) devise a method to approximately solve the resulting OF- k -SUP instance.

3.3 Growing a proper center subset

Next, we present the method that iteratively grows the desired proper center subset Λ (initialized as the empty set). In each iteration, the key idea is to first compute an independent center set Ψ on the remaining points $U = \Phi \setminus B(\Lambda, (1 + \sqrt{3})r^*)$ while ignoring the group membership, and then select a subset of Ψ to expand Λ . Specifically, for each group l , we choose $\Psi_l = \Psi \cap S_l$ with $|\Psi_l| \leq k_l - |\Lambda_l|$ to grow Λ . We repeat these iterations until U

Algorithm 1: Construction of the proper center subset

Input: The optimal radius r^* , and two λ -independent center set Γ_1 and Γ_2 obtained from the stream for $\lambda = \sqrt{3}r^*$.
Output: A set of centers Λ .

```

1 Initialization: Set  $\Phi \leftarrow \Gamma_1 \cup \Gamma_2$  and  $\Lambda \leftarrow \emptyset$ ;
2 while  $\Phi_l \neq \emptyset$  for each  $l \in \{1, 2\}$  do
3   if there exists two points  $c_1 \in \Gamma_1$  and  $c_2 \in \Gamma_2$  that covers
     all points of  $\Phi$  then
4     return  $\Lambda \leftarrow \Lambda \cup \{c_1, c_2\}$ ;
    //  $\sigma(S, \lambda)$  is  $\lambda$ -independent center set of  $S$  wrt  $\lambda$ 
5   Compute  $\Psi \leftarrow \sigma(\Phi, (1 + \sqrt{3})r^*)$  where  $\Psi$  contains at
     least one point of each  $\Gamma_l$ ;
6   Divide  $\Psi$  into  $\Psi_1 \subseteq \Gamma_1$  and  $\Psi_2 \subseteq \Gamma_2$ ; // Similarly,  $\Lambda$  divide
     into  $\Lambda_1$  and  $\Lambda_2$ 
7   if  $|\Psi_1| \leq k_1 - |\Lambda_1|$  and  $|\Psi_2| \leq k_2 - |\Lambda_2|$  then
8     return  $\Lambda \leftarrow \Lambda \cup \Psi$ .
9   else
10    Find  $l$  with  $|\Psi_l| < k_l - |\Lambda_l|$ ; // Note that there must
      exist such  $l$  in this case
11    Set  $\Lambda \leftarrow \Lambda \cup \Psi_l$  and update
       $\Phi \leftarrow \Phi \setminus B_\Phi(\Psi_l, (1 + \sqrt{3})r^*)$ .
12 return  $\Lambda$ .
```

contains points from only a single group (e.g., solely from Γ_1). For the case $m = 2$, the procedure is detailed in Alg. 1.

Lemma 4. Let $O = \{O_1, \dots, O_k\}$ be an optimal solution. During Alg. 1, any pair of points in Λ belongs to different clusters of O .

PROOF. According to Alg. 1, the distance between any pair of points in Λ is larger than $(1 + \sqrt{3})r^*$ (and thus larger than $2r^*$). Hence, any pair of points in Λ can not belong to the same cluster of O , since the diameter of any cluster in O is at most $2r^*$. $\square$

Lemma 5. When Alg. 1 terminates, one of the following conditions must hold:

- (1) Λ is already a feasible center set if Alg. 1 terminates at Step 4 or Step 8;
- (2) The returned Λ is a proper center subset of Φ .

The detailed proof is postponed to Appendix B.2.

3.4 Transformation to OF- k -SUP

Without loss of generality, assume that Γ_2 is completely covered by the returned proper center subset Λ , while a subset $U \subseteq \Gamma_1$ remains uncovered after executing Alg. 1. We then reduce the problem to a special case of OF- k -SUP as follows.

Definition 4. (Transformed OF- k -SUP instance) Let $U = \Gamma_1 \setminus B(\Lambda, (1 + \sqrt{3})r^*)$ for the set Λ returned by Alg. 1. Given budgets $k^U = k - |\Lambda|$ and $k_1^U = k_1 - |\Lambda_1|$, the goal is to construct a center set $C \subseteq \Phi$ that covers all points of U within radius $(1 + \sqrt{3})r^*$, subject to $|C_1| \leq k_1^U$ and $|C| \leq k^U$.

By Lem. 5, Λ is a proper center subset. Then, by Lem. 3, we immediately obtain the following corollary.

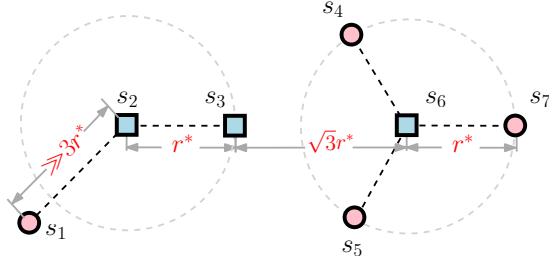


Figure 1: An example demonstrating that the optimal radius for the set of stored points can be up to $(1 + \sqrt{3})r^*$.

Corollary 2. For any fair k -center instance with optimal radius r^* , the transformed OF- k -SUP instance as in Def. 4 is feasible regarding radius $(1 + \sqrt{3})r^*$.

Therefore, it suffices to solve the resulting OF- k -SUP instance. In particular, when $k_1^U = k^U$, the problem further reduces to the classical k -supplier problem: use at most k^U centers from Φ to cover all points of U within radius $(1 + \sqrt{3})r^*$. However, even in this special case, the elegant $(1 + \sqrt{3})$ -approximation of [45] is inapplicable here, as it targets the standard k -supplier problem without consider group membership. Moreover, even if one could transfer their guarantee, the best achievable bound would be $(1 + \sqrt{3})^2 \approx 7.46$, because the optimum radius of the transformed instance can be as large as $(1 + \sqrt{3})r^*$. In Fig. 1, we demonstrate a tight example. Let S be partitioned into two groups of points $S_1 = \{s_1, s_4, s_5, s_7\}$ (pink circle) and $S_2 = \{s_2, s_3, s_6\}$ (blue square). For $k_1 = 1$ and $k_2 = 2$, the optimal solution to fair k -center is $\{s_1, s_2, s_6\}$ with radius r^* . Note that the distance between any pair of points in $\{s_4, s_5, s_7\}$ is $\sqrt{3}r^*$. After streaming, the stored λ -independent center sets are $\Gamma_1 = \{s_1, s_7\}$ and $\Gamma_2 = \{s_3\}$, and $\Phi = \Gamma_1 \cup \Gamma_2 = \{s_1, s_3, s_7\}$ for $\lambda = \sqrt{3}r^*$. Then, the optimal solution on Φ is $\{s_1, s_3\}$ with radius $d(s_3, s_7) = (1 + \sqrt{3})r^*$. Note that, $d(s_3, s_7)$ is greater than r^* the optimal radius on the original data set S .

So, instead of directly using the approximation of [45], we develop Alg. 2 that is tailored for the transformed problem, and ultimately obtain an approximation ratio of $1 + 2\sqrt{3} \approx 4.46$.

4 Approximating the OF- k -SUP problem

By setting $P = \Phi \setminus U$ for U and Φ produced by Alg. 1, we obtain a special case of OF- k -SUP as in Def. 4 that aims to cover all points in U using at most $k = k^U$ centers from $U \cup P$, with at most $\tau = k_1^U$ of them chosen from U . The key idea of our algorithm is to select the actual centers from $U \cup P$ by employing maximum matching for general graphs together with computational geometry techniques. More precisely, we transform our problem into a constrained maximum matching problem in an auxiliary graph $H(U, E)$ based on Lem. 1. For better readability, we use $\Gamma = U$ to emphasize the fact that U is a λ -independent set for S with $\lambda = \sqrt{3}r^*$. The auxiliary graph is defined as follows:

Definition 5. For the auxiliary graph $H(\Gamma, E)$, we include an edge $(u, u') \in E$ between each pair $u, u' \in \Gamma$ if and only if $d(u, u') \leq (1 + \sqrt{3})r^*$. Moreover, $E' \subseteq E$ is as follows: for edge $(u, u') \in E$, if

u and u' can be simultaneously covered by some point $v \in P$ with radius $(1 + \sqrt{3})r^*$, then we include edge (u, u') in E' .

By Lem. 1, assume there are h ($\leq k$) clusters in the optimal solution, each containing two vertices of Γ , with h_1 of these clusters centered at points in $\Gamma = U$ and h_2 centered at points in $P = \Phi \setminus U$ ($h_1 + h_2 = h$). Then we have the following lemma:

Lemma 6. There exists a matching M^* with $M'^* = M^* \cap E'$ such that both $|M^*| \geq h$ and $|M'^*| \geq h_2$ hold.

PROOF. We will show that the optimal solution induces a matching M^* (with $M'^* = M^* \cap E'$ satisfying $|M'^*| \geq h_2$). If two points $u, u' \in U$ belong to the same optimal cluster, then $d(u, u') \leq 2r^*$; hence edge $(u, u') \in E$ is added by the definition of $H(\Gamma, E)$. Moreover, by Lem. 1, the h edges corresponding to the h clusters (one edge per cluster) are vertex-disjoint and thus form a matching of size h . Among these edges, exactly h_2 edges belong to E' (those whose clusters are centered at points outside U). Therefore, $|M'^*| \geq h_2$ holds. $\square$

Thus, by Lem. 6, it suffices to find a maximum matching M in the auxiliary graph $H(\Gamma, E)$ whose number of edges in E' is at least h_2 . Unfortunately, this constrained maximum-matching problem is NP-hard, as it reduces to the exact matching problem [47]. Therefore, we design an algorithm tailored to our specific graph structure and objective. In essence, we develop a method to construct a set Z with the following properties:

Definition 6. The set Z is a comparable center set if and only if: (1) Z covers at least $|Z| + |M^*|$ points of Γ ; and (2) $|Z| \leq |M^*|$ and $|Z \cap \Gamma| \leq \tau$.

Lemma 7. If every point of Γ has neighbors in P , then any comparable center set Z yields a feasible center set.

PROOF. We construct a center set C according to Z . Initially, set $C = Z$. Let $\Gamma' = \Gamma \setminus B(Z, (1 + \sqrt{3})r^*)$ be the set of remaining uncovered points. We add $\tau - |C \cap \Gamma|$ points from Γ' to C as centers (chosen arbitrarily) and choose a set $P' \subseteq P$ that covers the remaining points of Γ' . By construction, we clearly have $|C \cap \Gamma| = |Z \cap \Gamma| + (\tau - |Z \cap \Gamma|) = \tau$. It remains to show $|C| \leq k$. The size of C is bounded by

$$\begin{aligned} |Z| + |\Gamma'| &= |Z| + (|\Gamma| - |Z| - |M^*|) \\ &= |\Gamma| - |M^*|, \end{aligned}$$

where the first equality is because Z covers at least $|Z| + |M^*|$ points of Γ by definition. Since $|\Gamma| - |M^*|$ equals the number of centers in the optimal solution, this completes the proof. $\square$

However, the set $\Gamma = U$ produced by Alg. 1 does not satisfy the condition of Lem. 7, i.e., there exist points in Γ without neighbors in P . So we introduce an additional algorithm to further reduce Γ such that it satisfies Lem. 7 as in Appendix B.4.

Following Lem. 7, our algorithm proceeds in two phases: (1) Find the centers for the subset of Γ that has no neighbors in P in the auxiliary graph H ; (2) Construct a comparable center set Z in the remaining graph of H and expand the center set accordingly.

To construct Z , we first compute a maximum matching M' in E' , and then compute the second maximum matching M'' in

Algorithm 2: Approximating OF- k -Sup

Input: $\Gamma (= U)$, P , the optimal radius r^* , two non-negative integers k , and τ ;
Output: The final center set C .

- 1 Set $Z, C \leftarrow \emptyset$ and $\Gamma' \leftarrow \Gamma \setminus B(P, (1 + \sqrt{3})r^*)$;
- 2 Compute $\Delta \leftarrow \sigma(\Gamma', 2r^*)$, and $\Gamma \leftarrow \Gamma \setminus B(\Delta, (1 + \sqrt{3})r^*)$;
 // Cover Γ by points of $\Gamma \cup P$ with maximum matching.
- 3 Construct the auxiliary graph $H(\Gamma, E)$ along with the subset $E' \subseteq E$ as in Def. 5;
- 4 Find a maximum matching M' in $H(\Gamma, E')$ by employing Edmonds' blossom algorithm [Micali and Vazirani](#);
- 5 Find the second maximum matching M'' in $H(\Gamma \setminus V(E'), E \setminus E')$;
- 6 Find the set of points $\Delta \subseteq \Gamma$ in which each point belongs to neither M' nor M'' ;
 /* Cover the points of Δ using centers for M'' and then M' . */
- 7 **while** $|Z \cap \Gamma| < \tau - |\Delta|$ **and** M'' has neighbors in Δ **do**
 8 Call Cover_Delta (M'' , Δ , H);
- 9 **while** $|Z \cap \Gamma| < \tau - |\Delta|$ **and** M' has neighbors in Δ **do**
 10 Call Cover_Delta (M' , Δ);
- 11 **if** $\Delta = \emptyset$ **then**
 12 Add $|M'|$ points of P and $|M''|$ points of Γ to Z , each as a center for an edge of M' or M'' , respectively;
- 13 **else**
 14 Add a subset of points of P into Z , each as a center for an edge of M' or a center for Δ .
- 15 **return** $C \leftarrow Z \cup \Delta$.
- 16 **Function** Cover_Delta (M, Δ, H) Find $u \in M$ that covers most points in Δ ;
- 17 Set $Z \leftarrow Z \cup \{u\}$, $M \leftarrow M \setminus e(u)$, and $\Delta \leftarrow \Delta \setminus N_H(u)$, where $e(u)$ is the edge containing u and $N_H(u)$ denotes the set of neighbor points of u in H .

$H(\Gamma \setminus E', E \setminus E')$. We then define Δ as the set of points not covered by $M' \cup M''$, and consequently find a point of M'' (and then M') that simultaneously covers an edge and at least one point of Δ in iterations until the desired set Z is constructed. The detailed algorithm is given in [Alg. 2](#). We first state a basic property of $M' \cup M''$:

Proposition 1. $M' \cup M''$ is a maximal matching in H , i.e., any edge of $H(\Gamma, E)$ is either in $M' \cup M''$ or is a neighbor of $M' \cup M''$.

Then, we have [Lem. 8](#), whose proof actually gives a method for computing a comparable center set.

Lemma 8. Let $\delta_M = |M^*| - |M' \cup M''|$. There exists a set $\Delta \subseteq \Gamma$ containing at least $2\delta_M$ points such that: (1) Each point of Δ is incident to neither M' nor M'' ; and (2) Each point of Δ is a neighbor of points in either M' or M'' .

PROOF. Let $\Delta_M = M^* - (M' \cup M'')$. Since $\delta_M = |M^*| - |M' \cup M''|$ by definition, we have that Δ contains δ_M points that are outside $M' \cup M''$. It remains to show that those points are either neighbors of M' or M'' . Suppose otherwise, i.e., there exists a point u that is a neighbor of neither M' nor M'' . Then, because u is incident to an

edge of $M^* \setminus (M' \cup M'')$, it has at least a neighbor u' that does not belong to M' or M'' . So edge (u, u') can be used to grow M' or M'' , which contradicts the fact that $M' \cup M''$ is maximal. $\square$

Lemma 9. The set Z produced by [Alg. 2](#) is a comparable center set.

The detailed proof of the above lemma is in [Appendix B.3](#). Eventually, we have the correctness of [Alg. 2](#) as below:

Theorem 1. [Alg. 2](#) produces a center set C satisfying: (1) $d(S, C) \leq (1 + 2\sqrt{3})r^*$; and (2) $|C| \leq k$ and $|C \cap S| \leq \tau$.

PROOF. First of all, by definition of Γ , we get that $d(\Gamma, S) \leq \sqrt{3}r^*$. Moreover, following [Alg. 2](#), we have $d(\Gamma, C) \leq (1 + \sqrt{3})r^*$. So we have $d(S, C) \leq d(\Gamma, C) + d(\Gamma, S) \leq (1 + 2\sqrt{3})r^*$.

Secondly, Δ is a proper center set. So after Step 2 of [Alg. 2](#), the points of Γ appear in $k - |\Delta|$ optimal clusters, among which at most $\tau - |\Delta|$ is centered at points of $\Gamma = U$. Moreover, at this step, each point of Γ has a neighbor in P . Then following [Lemma 7](#) and [9](#), C is a feasible center set, i.e., $|C| \leq k$ and $|C \cap S| \leq \tau$. $\square$

5 Experiments

We evaluate the performance of our methods on a variety of real-world and synthetic datasets, and report the clustering cost and running time to compare our algorithm with the baseline methods. More experimental details are provided in [Appendix C](#).

Datasets. We run the algorithms on five real-world datasets and two synthetic datasets: one with an optimal solution (*SyntA*) and another with a size of 100 GB (*SyntB*). More detailed information is provided in [Appendix C.1](#).

Algorithms. Compared with existing streaming fair k -center clustering algorithms, the baselines include:

- *two-pass 3-Approx*: a two-pass streaming 3-approximation algorithm proposed by Chiplunkar et al. [14];
- *7-Approx*: a one-pass streaming $(7+\epsilon)$ -approximation algorithm by Lin et al. [40];
- *5-Approx*: a recent one-pass streaming $(5 + \epsilon)$ -approximation algorithm by Guo et al. [26].

Moreover, our algorithm is a one-pass $(1 + 2\sqrt{3})$ -approximation algorithm (denoted as $(1 + 2\sqrt{3})$ -Approx or *OurAlg* in the experiments) and is further extended to handle the general case with m groups using a heuristic approach.

Metric. We measure the *utility* in terms of the cost of k -center clustering and running time (in seconds).

Experiment setup. In both real-world and simulated datasets, we configured the different parameters for both k and m to compare the average clustering cost and the maximum empirical approximation ratios of all algorithms, thereby evaluating the quality of our proposed method. To ensure fair center selection, we aligned the number of required centers from each group with the proportional size of that group. Following the fairness principle of disparate impact as outlined by Feldman et al. [19], we restricted the selection to at most k_l data points from the l th group to serve as centers. We then evaluated clustering quality and runtime across these datasets. All experiments were conducted on a Linux machine equipped with a 12th Gen Intel(R) Core(TM) i9-12900K CPU and 32 GB of RAM, using Python 3.8.

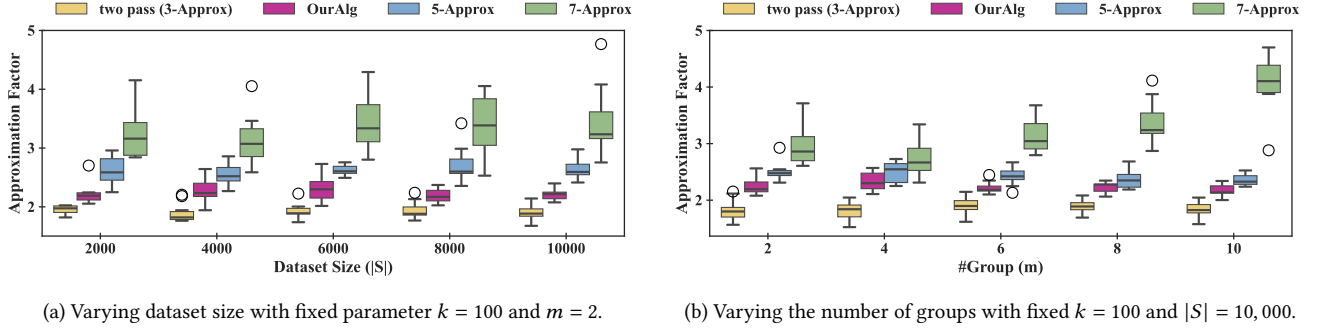
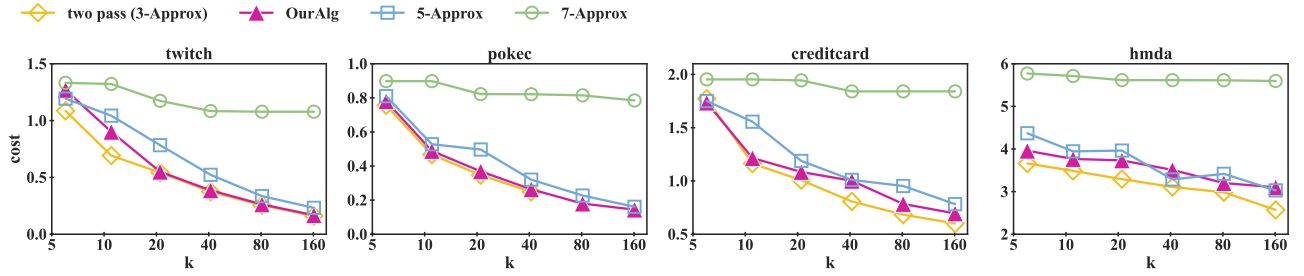


Figure 2: Comparison of empirical approximation ratio under different conditions.

Figure 3: Performance of streaming algorithms on different real-world datasets with varying parameter k .

5.1 Experiment results

In this section, we first analyze the experimental results on the synthetic dataset with an optimal solution under varying dataset sizes $|S|$ and different group sizes m . We then summarize the performance on real-world datasets and present the runtime analysis for both the real-world datasets and the 100 GB synthetic dataset.

Clustering quality on synthetic data. *SyntA* is a synthetic dataset constructed with an optimal solution for the fair k -center problem. Here, we define the maximal approximation factor as the *empirical approximation ratio*. The key observation is that *all experimental results on SyntA align with theoretical expectations: our algorithm outperforms both the one-pass 7-Approx and 5-Approx, while performing slightly worse than the two-pass 3-Approx algorithm.*

As shown in Fig. 2, under varying dataset sizes and group sizes, the clustering results closely follow the theoretical trends described above. In particular, both the average and maximum costs (empirical approximation ratios) align well with the expected theoretical order of the algorithms' approximation guarantees. In the one-pass setting, our algorithm demonstrates a clear advantage over the others by leveraging the properties of the Euclidean k -center to select more representative points while ensuring that each chosen point remains informative. It not only enables effective decision-making during the streaming process but also incorporates a post-processing procedure and analytical steps that reduce the actual center set to size k , thereby improving the overall clustering quality.

However, compared with the two-pass algorithm, our method lacks the opportunity to further refine the representative points

during a second pass, resulting in higher approximation factors. Nevertheless, it offers a substantial advantage in terms of computational efficiency, requiring significantly fewer resources than the two-pass streaming algorithm. It is also worth emphasizing the practical distinction between one-pass and two-pass algorithms. In real-world applications, a one-pass streaming algorithm does not need to store the entire dataset during processing, whereas a two-pass algorithm requires sufficient space to retain the data for the second pass.

On the other hand, in Fig. 2a, the clustering results are not affected by changes in the dataset size $|S|$, as clustering performance is typically influenced more by the data distribution than by the size of the dataset. In Fig. 2b, the main observation above remains consistent across different group size settings m , because all algorithms operate on independent center sets (see Def. 1) within each group, thereby ensuring equal chances for representation and enabling comparable results to be effectively handled during the post-processing stage or the second streaming pass.

Clustering quality on real-world data. In Fig. 3, we evaluate all algorithms on four real-world datasets, including large-scale social network data. The key trends observed in *SyntA* are similarly reflected in the real-world datasets, and the results remain consistent with the theoretical expectations. Next, the corresponding runtimes of these clustering results are presented in Fig. 4, and the evaluation of m -group fair k -center is illustrated in Appendix C.3.

As shown in Fig. 3, the clustering cost decreases for all algorithms as k increases. However, the 7-Approx algorithm exhibits a slower rate of decrease compared to the others, whose curves follow a more consistent trend. This behavior can be attributed to

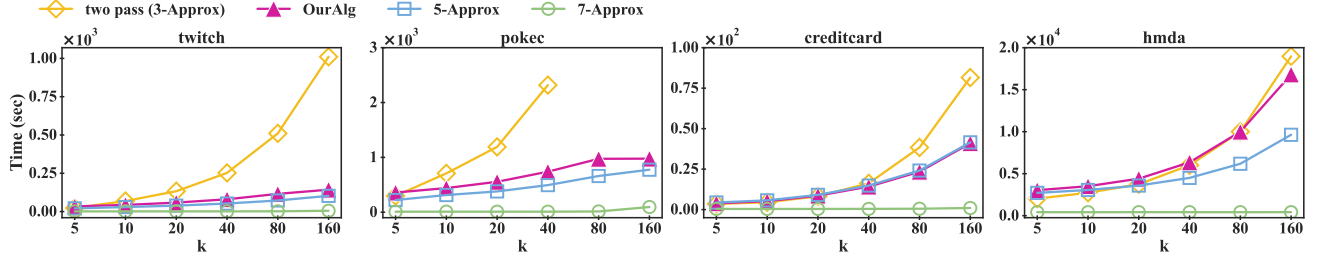


Figure 4: Running time of streaming algorithms on different real-world datasets with varying parameter k .

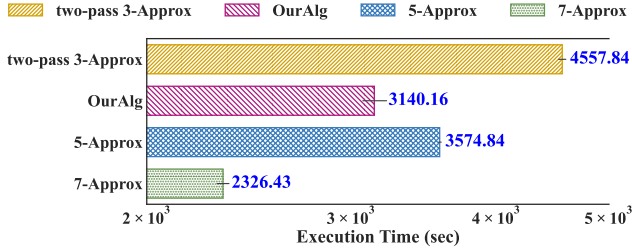


Figure 5: Runtime evaluation on 100G dataset (*syntB*).

the latter algorithms' ability to select higher-quality representative points as k grows, whereas 7-Approx struggles to identify suitable centers in the streaming setting. The other algorithms alleviate this limitation through their post-processing procedures. As k continues to increase, the marginal improvement in cost diminishes, since adding additional centers results in smaller incremental reductions in clustering cost.

OurAlg generally achieves performance between that of the two-pass 3-Approx and the 5-Approx algorithms, with the utility gap among them remaining stable across varying values of k . We attribute this to the use of independent center sets, which allows the algorithm to retain more representative points as k increases. However, in certain cases (around $k = 40$, *hmda* dataset), the results may be affected by the sequence of the streaming data.

Running time. In Fig. 4 (and Fig. 9 in Appendix C), we report the running times of the algorithms on real-world datasets under varying values of k for $m = 2$ (and $m = 4$). Consistent with the clustering cost results, we observe that the two-pass 3-Approx algorithm is the most time-consuming across all datasets, and its runtime increases as k grows. In contrast, our algorithm requires less running time than the two-pass 3-Approx, but slightly more than the other one-pass algorithms. We attribute this to the fact that our method achieves the lowest approximation ratio among the one-pass streaming algorithms, which takes a more detailed and complex selection process to ensure the representativeness of the chosen centers, thereby increasing computation time.

Meanwhile, the runtime advantage of our algorithm diminishes as the dataset size increases compared to the two-pass algorithm. This is because the I/O (data reading) process becomes relatively faster compared to the computational steps of the algorithm; as the algorithmic complexity grows, the additional processing time outweighs the gains from the I/O operations.

However, when we evaluate the running times of different algorithms on the 100 GB dataset (*SyntB*) in Fig. 5, the results differ from those observed on the real-world datasets. In this large-scale setting, our algorithm exhibits a faster runtime than the 5-Approx algorithm. We attribute this to the high dimensionality of *SyntB*. When dealing with complex data structures, the 5-Approx algorithm tends to perform more frequent operations to construct the auxiliary graph and compute distances, whereas our $1 + 2\sqrt{3}$ -Approx algorithm can handle most basic cases directly through Alg. 1. In addition, this experiment demonstrates that all the streaming algorithms are practical: they can operate efficiently under the streaming model and produce valid results even on datasets that exceed the available RAM capacity.

6 Conclusion

In this work, we present an improved streaming algorithm for Euclidean fair k -center that achieves an approximation ratio of 4.464, improving upon the previous best ratio of $5 + \epsilon$ for general metrics. After firstly caching sets of candidate representative centers from the data stream, we propose a novel method for selecting the final center set from these sets of candidate centers by reducing the problem to a special case of the outsourcing fair k -supplier problem and then developing an approximation algorithm accordingly. The key ingredients for the reduction and the approximation are our designated *proper center subset* and *comparable center set*, respectively. This result shows that streaming fair k -center admits a strictly better approximation ratio in Euclidean space than in general metrics. Competing with the theoretical results, we also show experimentally that our algorithm achieves better practical solution quality and runtime on real-world social network datasets than state-of-the-art algorithms. Since the best-known approximation ratio for the standard k -center problem is 2 in both Euclidean and general metrics, it remains a fundamental open problem whether offline Euclidean fair k -center admits an approximation ratio better than 3; that is, can it be approximated with a strictly better ratio than the state-of-the-art ratio of 3 in general metrics?

Acknowledgments

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APPENDIX

A Tight Example for Streaming Metric Fair k -Center

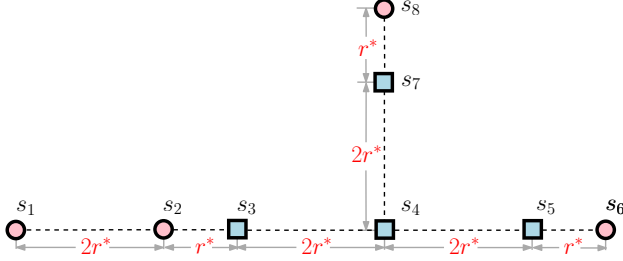


Figure 6: An example demonstrating the ratio 5 is tight in metric space.

In Fig. 6, the original dataset consists of two groups are $S_1 = \{s_1, s_2, s_6, s_8\}$ (pink points) and $S_2 = \{s_3, s_4, s_5, s_7\}$ (blue points), with fairness constraints $k_1 = 2$ and $k_2 = 3$. The optimal solution is $C^* = \{s_1, s_2, s_4, s_5, s_7\}$, achieving a cluster radius r^* . The stored points after Stage 1 of our algorithm can be $\Gamma_1 = \{s_2, s_6, s_8\}$ and $\Gamma_2 = \{s_4\}$. In contrast to the optimal solution, a possible solution selected from $\Gamma_1 \cup \Gamma_2$ by our algorithm is $C = \{s_4, s_6, s_8\}$, resulting in a distance of $d(s_1, C) = 5r^*$.

B Missing proofs and more detailed algorithms

B.1 Proof of Lem. 2

PROOF. Let O be the optimal center set. We can construct such a set C (initially empty) from Φ as follows:

For each center $o \in O$ belonging to group l do

Find $o' \in \Gamma_l$ with $d(o', o) \leq \sqrt{3}r^*$ and set $C \leftarrow C \cup \{o'\}$.

First, according to the rule of the streaming stage and Def. 1, such a point o' must exist in Γ_l .

We now verify the correctness of Condition (1). For any point $u \in \Phi$ whose center in the optimal clustering is o , let $o' \in C$ be the replacement of o . Then, we have $d(u, C) \leq d(u, o) + d(o', o) \leq (1 + \sqrt{3})r^*$ (as shown in Fig. 7).

Lastly, Condition (2) follows immediately from the construction, since C contains the same number of centers from each group as the optimal center set O . $\square$

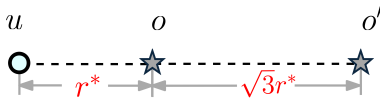


Figure 7: Demonstration that any point u can find in the $\sqrt{3}r^*$ -independent center set a center o' that shares the same group with its optimal center o , within distance bounded by $(1 + \sqrt{3})r^*$.

B.2 Proof of Lem. 5

PROOF. For Condition (1), if the algorithm terminates at Step 8, it is immediate that Λ is a feasible clustering solution.

Following Lem. 4, we have $|\Lambda \cup \Psi| \leq k$ throughout Alg. 1. So either the condition in Step 7 holds, or during the while loop we always have $|\Psi_l| > k_l - |\Lambda_l|$ and $|\Psi_{3-l}| < k_{3-l} - |\Lambda_{3-l}|$. Therefore, at the beginning of each while loop (before Step 4), we have both $k_l - |\Lambda_l| \geq 1$ and $k_{3-l} - |\Lambda_{3-l}| \geq 1$. Consequently, in Step 4, after adding c_1, c_2 , the set Λ remains a feasible center set that satisfies: i) $|\Lambda| \leq k$ and $|\Lambda_l| \leq k_l$ for $l \in 1, 2$; ii) $d(\Lambda, \Phi) \leq (1 + \sqrt{3})r^*$.

For Condition (2), we show that when Alg. 1 terminates, i) the remaining points in Φ belong to at most $k - |\Lambda|$ clusters of the optimal solution O ; ii) this clusters can be covered, within radius $(1 + \sqrt{3})r^*$, by at most $k_l - |\Lambda_l|$ centers from group l , for each l . We prove this by induction over the iterations. Let Λ^j denote Λ at the beginning of iteration j . Suppose that in iteration j , the residual set Φ can be covered using $k_l - |\Lambda_l^j|$ and $k_{3-l} - |\Lambda_{3-l}^j|$ centers from group l and $3 - l$, respectively. Suppose that in iteration j we add Ψ_{3-l} to Λ . It remains only to show $\Phi' = \Phi \setminus B(\Psi_{3-l}, (1 + \sqrt{3})r^*)$ can be covered by $k_l - |\Lambda_l^j|$ from group l and $k_{3-l} - |\Lambda_{3-l}^j| - |\Psi_{3-l}|$ centers from group $3 - l$.

By Lem. 4, the points in $\Psi_{3-l} \cup \Lambda^j$ must lie in distinct optimal clusters. In the clustering witnessing the inductive hypothesis for iteration j , let o_{3-l}^1 and o_{3-l}^2 be the numbers of clusters intersecting Ψ_{3-l} whose centers belong to groups 1 and 2 by hypothesis, respectively (so $o_{3-l}^1 + o_{3-l}^2 = |\Psi_{3-l}|$). Removing $B(\Psi_{3-l}, (1 + \sqrt{3})r^*)$ eliminates these $|\Psi_{3-l}|$ clusters from Φ , hence $\Phi' = \Phi \setminus B(\Psi_{3-l}, (1 + \sqrt{3})r^*)$ can be covered by $k_{3-l} - |\Lambda_{3-l}^j| - o_{3-l}^{3-l}$ centers from group $(3 - l)$, along with $k_l - |\Lambda_l^j| - o_{3-l}^l$ centers from group l .

Since we add Ψ_{3-l} to Λ , we have already fixed $|\Psi_{3-l}|$ centers from group $(3 - l)$, including the o_{3-l}^l clusters previously centered at points of group l in the hypothesis. Therefore, among the $k_{3-l} - |\Lambda_{3-l}^j| - o_{3-l}^{3-l}$ clusters of Φ' previously assigned to group $(3 - l)$, we select $k_{3-l} - |\Lambda_{3-l}^j| - |\Psi_{3-l}|$ to remain covered by group $(3 - l)$. The remaining clusters in Φ' (at most o_{3-l}^l) are then covered by group l (changing from group $3 - l$). Note that, this change is feasible because Ψ provides centers from group l for all those clusters. This completes the proof. $\square$

B.3 Proof of Lem. 9

PROOF. Let M^* be the matching according to the optimal solution with $|M^*| = h$ and $|M'^*| = h_2$. We consider the following two cases and complete the proof:

- $\Delta = \emptyset$ in Step 11 of Alg. 2.

In this case, Z covers all points of M^* , whose number is $2|M^*|$. On the other hand, $|Z| = |M| \leq |M^*|$. So Z covers at least $|Z| + |M^*|$ points.

- Otherwise.

As M' is the maximum matching in E' , we surely have $|M'| \geq |M'^*| = h_2$. On the other hand, we have $|M''| \leq h_1$.

We further consider two possibilities in this case:

Algorithm 3: Further growing of Λ and reducing of Φ

Input: Two λ -independent center set Γ_1 and Γ_2 where Γ_{3-l} has uncovered points while Γ_l is completely covered; the optimal radius r^* ; a proper center subset Λ output by Alg. 1.

Output: Center set Λ' grown from Λ .

- 1 Set $U \leftarrow \Gamma_{3-l} \setminus B(\Lambda, (1 + \sqrt{3})r^*)$, $\Lambda' \leftarrow \Lambda$, and $V \leftarrow \Gamma_l \setminus \Lambda_l$;
/* Note that we need only to cover U the uncovered points of Γ_{3-l} , possibly using points of $V \subseteq S_l$ as new centers if $|\Lambda_l| < k_l$. */
- 2 Construct a simple graph $G(U \cup V; E)$ with $e(u, v) \in E$ if and only if $d(u, v) \leq (1 + \sqrt{3})r^*$ for $u \in U, v \in U \cup V$;
/* Phase I-2: Select centers from U' that is the set of points of U that do not neighbor any points of V */
- 3 Set $U' \leftarrow U \setminus N(V)$;
- 4 Compute $C_{U'} \leftarrow \sigma(U', (1 + \sqrt{3})r^*)$ and set $U \leftarrow U \setminus B(C_{U'}, (1 + \sqrt{3})r^*)$;
- 5 Set $\Lambda' \leftarrow \Lambda \cup C_{U'}$;
- 6 Return Λ' .

– $|Z \cap \Gamma| = \tau - |\Lambda|$.

In this case, there are at least $\tau - |\Lambda| - |M2|$ points in Z each of which covers at least three points. Thus, the number of points we use Z to cover is

$$\begin{aligned} & 2|M| + (\tau - |\Lambda| - |M2|) \\ &= |M| + |M'| + |M''| + (\tau - |\Lambda| - |M2|) \\ &\geq |M| + |M'| + |M''| + (h_2 - |M2|) \\ &= |Z| + |M^*| \end{aligned}$$

where the first inequality is because of $h_2 \leq \tau - |\Lambda|$, the second equality is because of $|Z| = |M|$.

- Otherwise, the remaining edges in $M' \cup M''$ have no neighbors in Δ . Then, because every point of Δ neighbors $M' \cup M''$ and the algorithm always finds u that covers most points of Δ by Step 17, we have that Z covers at least δ_M points of Δ . That is, the number of points Z covers is

$$2|M| + \delta_M = 2|M| + (|M^*| - |M|) = |Z| + |M^*|.$$

By combining the above cases, we complete the proof. $\square$

B.4 An additional algorithm

In addition to Alg. 1, we provide Alg. 3, a follow-up algorithm that further reduces U , the set of remaining points of Φ , such that every point in the further processed U has neighbors in $\Phi \setminus \Lambda$. Then, the property of Lem. 7 applies.

Lemma 10. *The set Λ' produced by Alg. 3 remains a proper center subset with respect to Φ .*

PROOF. We will first show $C_{U'}$ is a proper center subset with respect to U . By Lem. 5, the points of U appear in at most $k - |\Lambda|$ optimal clusters, and at most $k_h - |\Lambda_h|$ for each $h \in \{1, 2\}$.

Because the points of U' have no neighbors V , they cannot lie in optimal clusters centered at points from group l . On the other hand, any pair of points in $C_{U'}$ does not belong to the same optimal cluster. Therefore, the set U (before Step 5 of Alg. 3) appears in at most $k - |\Lambda| - |C_{U'}|$ optimal clusters, with at most $k_l - |\Lambda_l|$ centers from group l and at most $k_{3-l} - |\Lambda_{3-l}| - |C_{U'}|$ centers from group $3 - l$. $\square$

C Additional experimental analysis

We summarize all the datasets' statistical information in Table 1.

C.1 Dataset

We run the algorithms on five real-world datasets, including social network datasets, and two synthetic datasets: one with an optimal solution (SyntA) and another with a size of 100 GB (SyntB).

Real-world datasets. Following the recent related works [7, 21], we evaluate the utility of clustering on five real-world datasets as follows.

- *Students' Social Network Profile Clustering dataset (sns)*¹ This dataset contains 30,000 high school students who maintained social networking profiles between 2006 and 2009. It was compiled by Brett Lantz during his research on teenage identities at the University of Notre Dame and later used as an example in his book [39]. We use the 26 feature for each record, and the gender of individuals is used as the clustering sensitive attribute for two protected groups: male and female.
- *twitch dataset*² [48] contains 168K data points, each representing a user of Twitch, a live-streaming platform for gamers. We use the sensitive attribute "mature" to define the groups, which takes binary values of 0 and 1.
- *pokec dataset*³ [50] contains 1.6M records of individuals from one of the largest social networks in Slovakia. We use the fixed fileable attributes for the user to measure the utility and clean the data format attributes. Gender (i.e., female and male) and age (i.e., every twenty years between twenty and sixty, and the other two fields out of this range) with $m = 4$ are considered sensitive attributes.
- *credit-card dataset*⁴: [55] contains 30K data points, each referring to a credit card holder from Taiwan. We use the 23 features for each record, and sex of individuals is used as the clustering sensitive attribute to generate two protected groups: male and female.
- *hmda dataset*⁵ comprises around 12M records of loan-level information about mortgages in the US market in 2024, and lien_status ($m = 2$) is considered as the sensitive attribute in our work.

For all datasets, we apply MaxScale normalization to all columns, standardize date-format attributes, and remove records with missing values.

¹<https://github.com/NikhilSalv/SNS-Data-Analysis-with-K-Means-Clustering/blob/main/snsdata.csv>

²https://snap.stanford.edu/data/twitch_gamers.html

³<http://snap.stanford.edu/data/soc-Pokec.html>

⁴<https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>

⁵<https://www.consumerfinance.gov/data-research/hmda/>

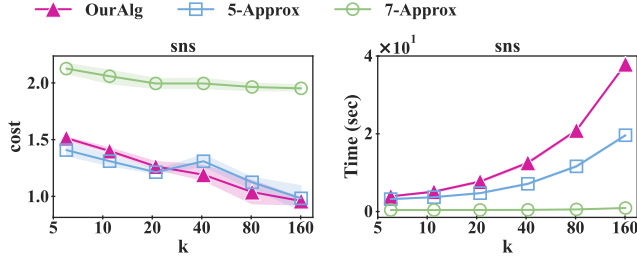


Figure 8: Comparison of our algorithm with one-pass streaming fair k -center baselines on the *sns* dataset.

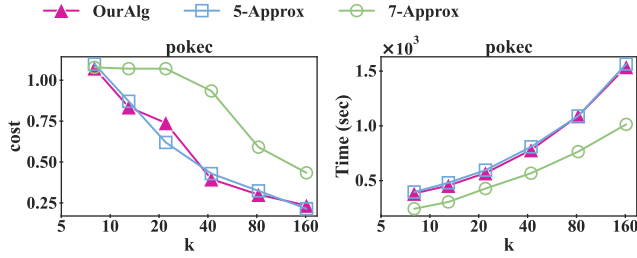


Figure 9: Comparison of different one-pass streaming fair k -center algorithms on the *pokec* dataset with the protected attribute *age* ($m = 4$).

Table 1: Datasets Statistic.

Dataset	#Record	#Dim.	Feature	#Group Record
sns	24,005	26	gender	[19311,4694]
creditcard	30,000	22	sex	[18112,11888]
twitch	168,114	5	mature	[89081,79033]
pokec	1,632,640	4	gender	[828304,804336]
pokec	1,632,640	4	age	[865742, 685664, 75973, 5261]
hmda	12,236,251	6	lien_status	[9381260,2854991]
SyntA	[2k–10k]	2	[2,4,6,8,10]	-
SyntB	4,000,000	1,000	$m = 2$	[89081,79033]

Synthetic datasets. We provide two synthetic datasets with fairness that serve as supplementary datasets for special cases:

- *SyntA*: the simulated dataset provides an exact optimal solution for fair k -center following [38] to assess the *empirical approximation ratio*.
- *SyntB*: the large-scale *100G* dataset includes more time for file I/O operations inspired by [14], making it ideal for measuring computational efficiency.

To reduce randomness, we generate each dataset (SyntA) at least five times with different fairness constraints for evaluation under various settings.

C.2 More Experimental Results

In Fig. 8, we report fair clustering results on the high-dimensional *sns* dataset, which has imbalanced groups. We observe that the

clustering cost decreases steadily as k increases, and the results improve for larger k . We attribute this to the dataset imbalance: as the number of k grows, our algorithm has more opportunities than 5-APPROX to select candidate centers from the smaller groups, leading to better fairness-aware coverage. However, our algorithm often requires more runtime than the other two algorithms on the *sns* dataset. For the same reason, the imbalanced group sizes and high-dimensional vectors increase the size of our candidate set compared to the baselines, which in turn increases the time spent computing distances.

C.3 Extension to m group

For the dataset with a general number of m groups, we evaluate algorithm performance and runtime on the social network dataset *pokec* under varying values of k , with m set to 4, as shown in Fig. 9. Compared with the 5-Approx and 7-Approx algorithms, our method generally achieves the lowest cost values across different k , aligning well with theoretical expectations. Moreover, when $m = 4$, the three algorithms exhibit similar utility for small k , whereas a more noticeable gap appears when $m = 2$ in Fig. 3. We attribute this to the fact that increasing m enables the algorithm to retain a more diverse set of points from the stream, allowing the 5-Approx and 7-Approx algorithms to more effectively select representative centers. For the same reason, the 7-Approx algorithm exhibits a more pronounced decrease in utility as k increases in Fig. 9.

In addition, in Fig. 3 and Fig. 9, the results of 5-Approx are not significantly improved by *OurAlg*, though *OurAlg* slightly outperforms 5-Approx. We believe this is due not only to the influence of data ordering within the stream, but also to the fact that, although the 5-Approx algorithm is designed for general metric spaces, all our experiments were conducted using the Euclidean distance. Consequently, it is reasonable that the two algorithms yield similar clustering results. Furthermore, the runtime results in Fig. 9 show that both algorithms require comparable computation time during post-processing, suggesting that they exhibit similar post-processing selection behavior on this dataset when $m = 4$.

Moreover, when applying the algorithm to social network datasets with biased features (e.g., Students' Social Network Profile Clustering dataset *sns* with gender as a fair feature), we observe that the advantages of *OurAlg* using the Euclidean distance become less pronounced for smaller values of k . We attribute this to the distance concentration phenomenon in high-dimensional spaces and the influence of data characteristics, where biased features act as fairness constraints (see Table 1). In high-dimensional settings, pairwise distances tend to converge, rendering geometric intuition less effective and diminishing the benefits of the Euclidean metric. Moreover, since the 5-Approx algorithm also relies on the Euclidean distance in the experiment, its clustering results are similar to our algorithm is reasonable. In addition, the presence of biased features makes the selection of representative centers more challenging; however, as the number of centers increases, more points can be retained during the streaming process, which alleviates this difficulty for our algorithm.